



THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, (A
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ELC REPORT

HDL Implementation of Digital Clock

Group Number – Group 13

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Introduction:

A digital clock is a sequential digital circuit used to display time in hours, minutes, and seconds. In this activity, a **24-hour digital clock** was designed and implemented using **Verilog HDL**. The design uses a clock signal and counters to update seconds, minutes, and hours. A reset input initializes the clock to **00:00:00**, and the time automatically changes from seconds to minutes and minutes to hours, with the clock resetting to **00:00:00 after 23:59:59**. The design was verified using a Verilog testbench through simulation.

Software Used: Xilinx Vivado (Verilog HDL Simulation)

Verilog Code:

```
module digital_clock
(
    input wire clk,
    input wire reset,
    output reg [4:0] hours,
    output reg [5:0] minutes,
    output reg [5:0] seconds
);

always @(posedge clk or posedge reset) begin
    if (reset) begin
        hours <= 0;
        minutes <= 0;
        seconds <= 0;
    end
    else begin
        if (seconds == 59) begin
```

```
seconds <= 0;
```

```
if (minutes == 59) begin
```

```
    minutes <= 0;
```

```
    if (hours == 23)
```

```
        hours <= 0;
```

```
    else
```

```
        hours <= hours + 1;
```

```
    end
```

```
else begin
```

```
    minutes <= minutes + 1;
```

```
end
```

```
end
```

```
else begin
```

```
    seconds <= seconds + 1;
```

```
end
```

```
end
```

```
end
```

```
endmodule
```

Testbench Code for Simulation

```
`timescale 1ns/1ps

module digital_clock_tb;

    reg clk;

    reg reset;

    wire [4:0] hours;

    wire [5:0] minutes;

    wire [5:0] seconds;

    digital_clock uut (
        .clk(clk),
        .reset(reset),
        .hours(hours),
        .minutes(minutes),
        .seconds(seconds)
    );

    // Generate clock
    always #5 clk = ~clk;

    initial begin
        clk = 0;

        reset = 1;

        // Reset clock

        #10;

        reset = 0;

        // Run simulation

        #4000;

        $finish;
```

end

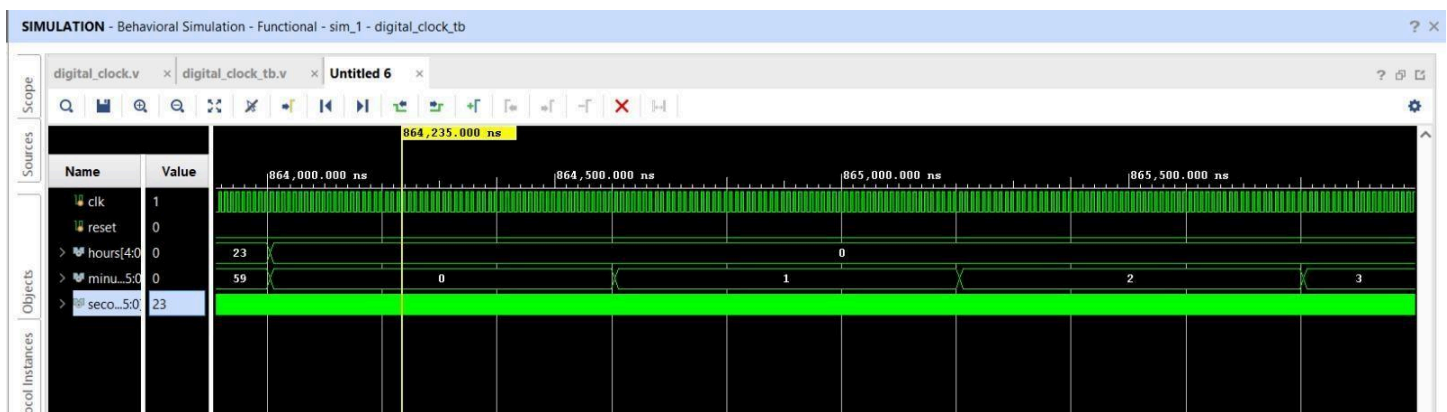
initial begin

```
$monitor("Time = %0d:%0d:%0d",  
        hours, minutes, seconds);
```

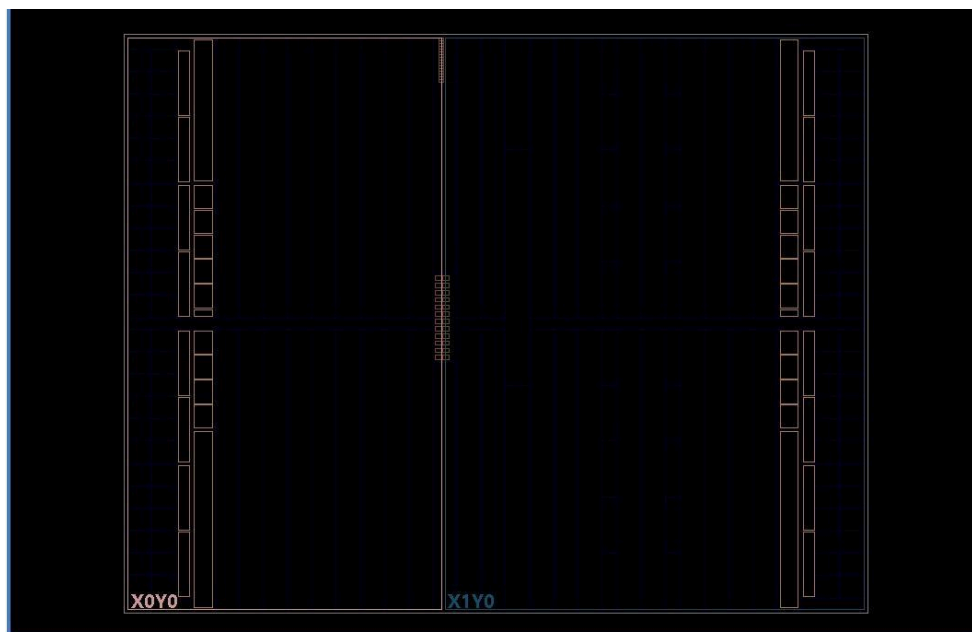
end

endmodule

Simulation Waveform



Synthesized Design



Conclusion:

The digital clock was successfully designed and implemented using Verilog HDL and synthesized in Xilinx Vivado. The design accurately generates and updates hours, minutes, and seconds according to the clock input, with a reset function to initialize the time. The synthesized design was also verified using a testbench, demonstrating the correct functionality of the digital clock.

Applications:

The digital clock can be used in **digital watches, alarm clocks, electronic appliances, embedded systems, and industrial timing systems** where accurate time measurement is required. It can also be integrated into **FPGA-based systems** for time-dependent operations and control applications. Additionally, this project is useful for educational purposes as it demonstrates the practical implementation of **Verilog HDL, counters, sequential logic, simulation, and FPGA synthesis**.